

machine Learning

Logistic Regression Model

Lec 4

Instructor:

Assoc Lecturer : Ahmed Yousry

CLASSIFICATION

The classification problem is just like the regression problem, except that the values y we now want to predict take on only a **small number of discrete values**.

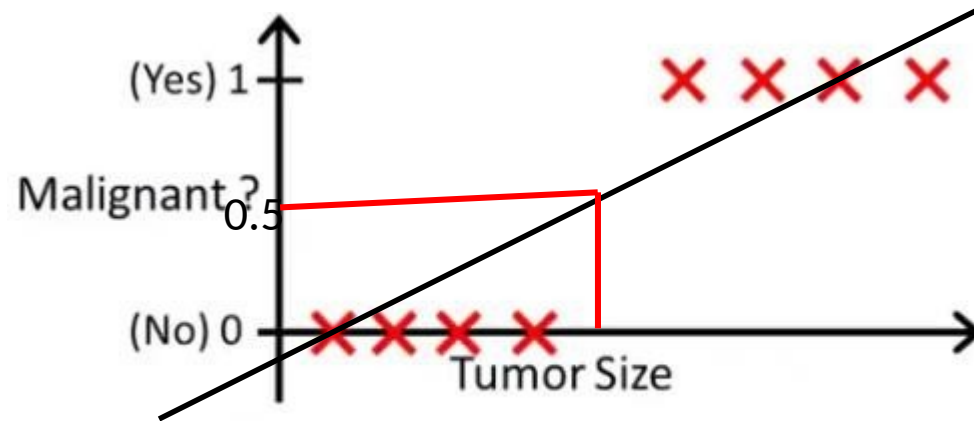
Some Example of Classification problem

- Email : Spam / Not spam
- Tumor: Malignant/ Benign

$$y \in \{0, 1\}$$

0: "Negative Class" (e.g., benign tumor)

1: "Positive Class" (e.g., malignant tumor)



Threshold classifier output $h_{\theta}(x)$ at 0.5:

If $h_{\theta}(x) \geq 0.5$, predict “ $y = 1$ ”

If $h_{\theta}(x) < 0.5$, predict “ $y = 0$ ”

DEFINITION

■ Binary Logistic Regression

- We have a set of feature vectors X with corresponding binary outputs

$$X = \{x_1, x_2, \dots, x_n\}^T$$
$$Y = \{y_1, y_2, \dots, y_n\}^T, \text{ where } y_i \in \{0, 1\}$$

- We want to model $p(y|x)$

$$p(y_i = 1 | x_i, \theta) = \frac{1}{1 + \exp(-x_i \theta)}$$

By definition $p(y_i = 1 | x_i, \theta) \in \{0, 1\}$. We want to transform the probability to remove the range restrictions, as $x_i \theta$ can take any real value.

USING ODDS

■ Odds

p : probability of an event occurring

$1 - p$: probability of the event not

occurring The odds for event i are then

defined as $odds_i = \frac{p_i}{1 - p_i}$

Taking the **log** of the odds removes the range restrictions.

$$\log_e \left(\frac{p_i}{1 - p_i} \right) = \sum_j \theta_j x_{ij} = X_i \theta$$

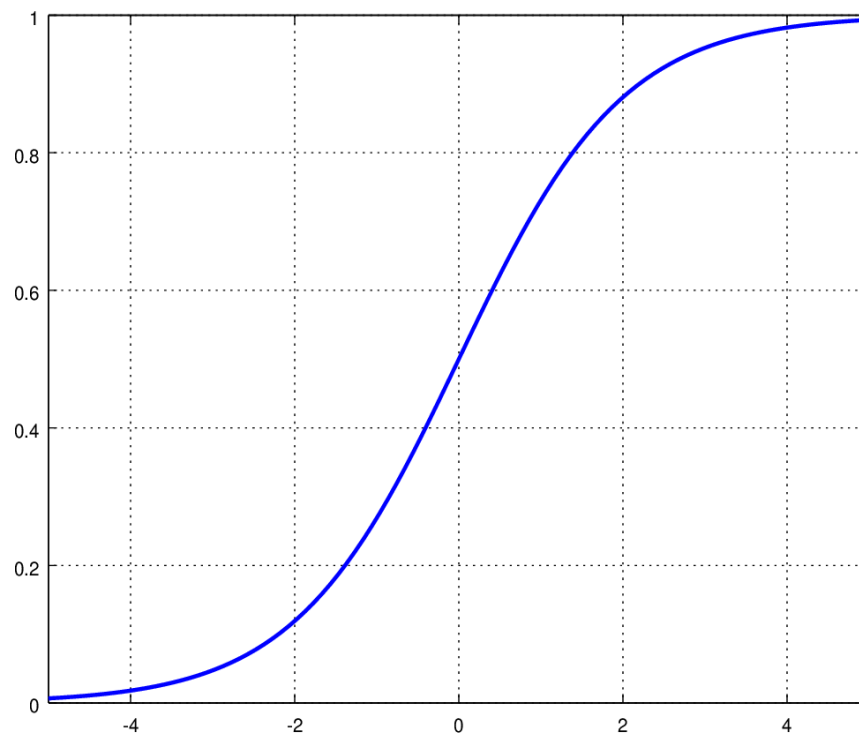
This way we map the probabilities from the $[0; 1]$ range to the entire number line (real value).

HYPOTHESIS FUNCTION

$$\log \left(\frac{p_i}{1 - p_i} \right) = x_i^T \theta$$

$$\frac{p_i}{1 - p_i} = e^{x_i^T \theta}$$

$$p_i = \frac{e^{x_i^T \theta}}{1 + e^{x_i^T \theta}}$$



Standard logistic sigmoid function

$$h_{\theta}(x) = \theta^T x$$

$$h_{\theta}(x) = g(\theta^T x)$$

$$p_i = g(\theta^T x) = \frac{1}{1 + e^{-\theta^T x}}$$

LOGISTIC REGRESSION MODEL

Linear Regression

$$h_{\theta}(x) = \theta^t x$$

Logistic Regression

$$g(\theta^t x) = \begin{cases} 1, & \frac{1}{1+e^{-\theta^t x}} \geq 0.5 \\ 0, & \frac{1}{1+e^{-\theta^t x}} < 0.5 \end{cases}$$

$$p(y_i = 1 | x_i, \theta) = \frac{1}{1 + e^{-\theta^t x}}$$

$$p(y_i = 0 | x_i, \theta) = 1 - \frac{1}{1 + e^{-\theta^t x}}$$

$$p(y_i | x_i : \theta) = \left(\frac{1}{1 + e^{-\theta^t x}} \right)^{y_i} \left(1 - \frac{1}{1 + e^{-\theta^t x}} \right)^{1-y_i}$$

Interpretation of Hypothesis Output

$h_{\theta}(x)$ = estimated probability that $y = 1$ on input x

Example: If $x = \begin{bmatrix} x_0 \\ x_1 \end{bmatrix} = \begin{bmatrix} 1 \\ \text{tumorSize} \end{bmatrix}$

$$h_{\theta}(x) = 0.7$$

Tell patient that 70% chance of tumor being malignant

$$h_{\theta}(x) = p(y = 1 | x; \theta) \quad \text{"probability that } y = 1, \text{ given } x, \text{ parameterized by } \theta"$$

$$P(y = 0 | x; \theta) + P(y = 1 | x; \theta) = 1$$
$$P(y = 0 | x; \theta) = 1 - P(y = 1 | x; \theta)$$

ESTIMATION OF COEFFICIENTS Θ

■ Maximum Likelihood Estimation (MLE)

1. Step 1 : get the probability for all observations

$$p(y \mid X : \theta) = \prod_{i=1}^m \left(\frac{1}{1 + e^{-\theta^t x}} \right)^{y_i} \left(1 - \frac{1}{1 + e^{-\theta^t x}} \right)^{1-y_i}$$

2. Step 2 : Express this as a function of θ , where X and y are fixed parameters

$$L(\theta) = p(y \mid X : \theta)$$

3. Step 3 : Maximize $L(\theta)$ likelihood function

$$l(\theta) = \prod_{i=1}^m \left(\frac{1}{1 + e^{-\theta^t x}} \right)^{y_i} \left(1 - \frac{1}{1 + e^{-\theta^t x}} \right)^{1-y_i}$$

We can simplify $L(\theta)$ by taking its **log** and then differentiate to get the gradient.

$$j(\theta) = l(\theta) = \sum_1^m \left[y_i \log \left(\frac{1}{1 + e^{-\theta^t x}} \right) + (1 - y_i) \log \left(1 - \frac{1}{1 + e^{-\theta^t x}} \right) \right]$$

$$\begin{aligned} \frac{d}{d\theta} j(\theta) &= \frac{d}{d\theta} \sum_1^m \left[y_i \log \left(\frac{1}{1 + e^{-\theta^t x}} \right) + (1 - y_i) \log \left(1 - \frac{1}{1 + e^{-\theta^t x}} \right) \right] \\ &= \sum_{i=1}^m \left(y_i - \frac{1}{1 + e^{-\theta^t x_i}} \right) x_i \end{aligned}$$

GRADIENT DESCENT

■ in Linear Regression

Want $\min_{\theta} J(\theta)$:

Repeat {

$$\theta_j := \theta_j - \alpha \frac{\partial}{\partial \theta_j} J(\theta)$$

}

(simultaneously update all θ_j)

■ in Logistic Regression

We can now use gradient ascent to maximize $j(\theta)$ The update rule will be:
repeat until convergence

{

$$\theta_j = \theta_j + \alpha \sum_{i=1}^m \left(y_i - \frac{1}{1 + e^{-\theta^t x_i}} \right) x_{ij}$$

}

Any Question